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APACHE AIRFLOW SETUP DOCUMENT

Docker Desktop Setup (Windows)

1. Download **Docker Desktop Installer (AMD64)** for Windows.
2. Run installer → accept license → finish setup.
3. **Restart** your system.
4. Open **Docker Desktop** from Start menu.
5. Now Docker is ready — we can deploy **Apache Airflow** with Docker.

Apache Airflow Setup (Windows)

Create a new file

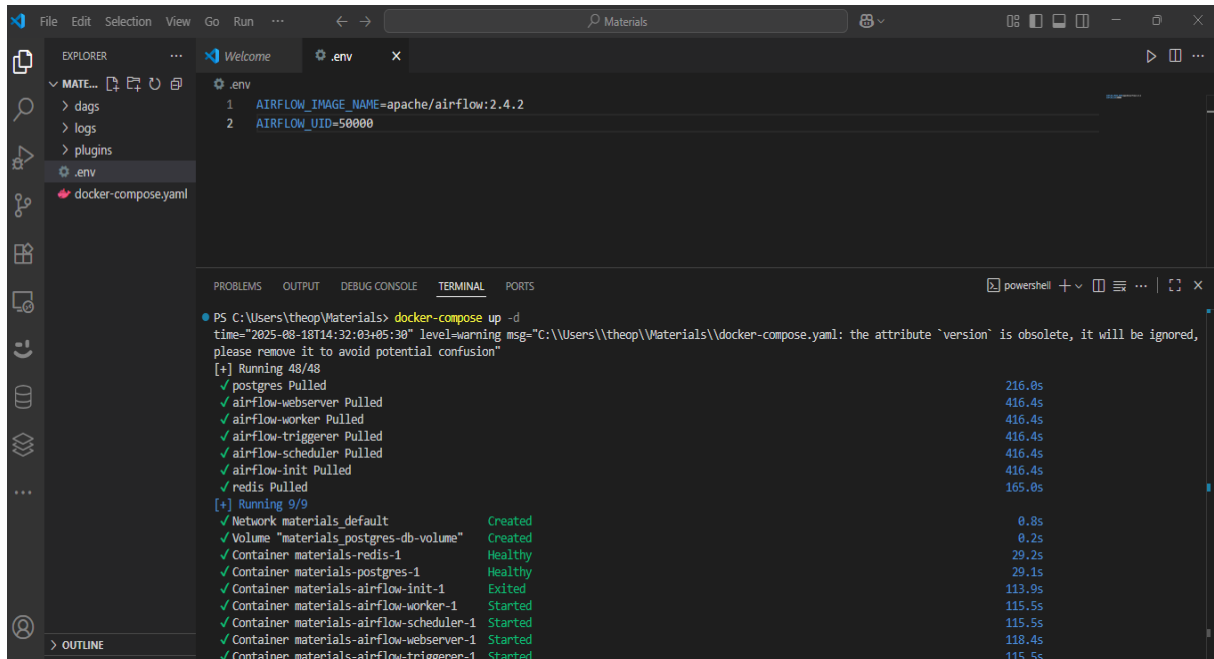
- In VS Code:
 - Right-click → New File → name it: **.env**
 - Open .env and paste this:

```
AIRFLOW_IMAGE_NAME=apache/airflow:2.4.2  
AIRFLOW_UID=50000
```

Save the file

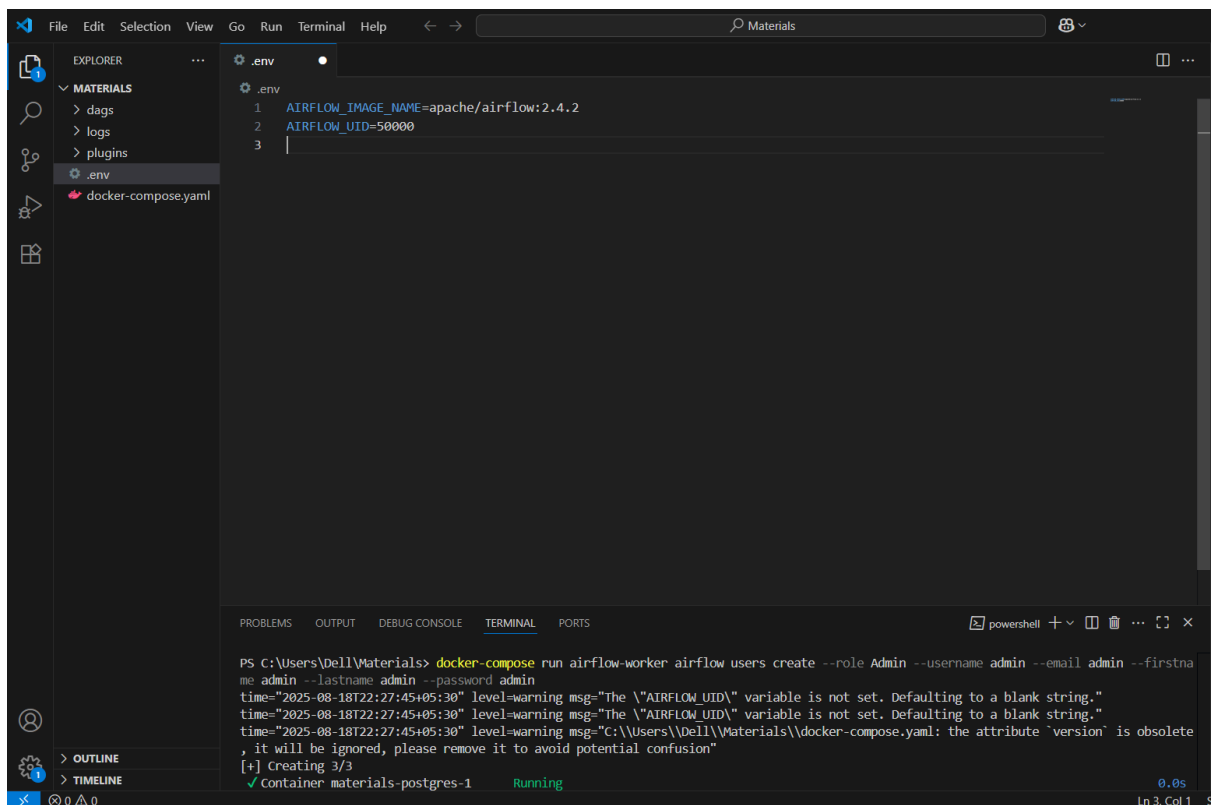
- Press **Ctrl+S** in VS Code (or File → Save).
- Ensure the file is in the **same folder** as **docker-compose.yaml**

Start Airflow Services



```
PS C:\Users\theop\Materials> docker-compose up -d
time="2025-08-18T14:32:03+05:30" level=warning msg="C:\Users\theop\Materials\docker-compose.yaml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 48/48
  ✓ postgres Pulled                                216.0s
  ✓ airflow-webserver Pulled                        416.4s
  ✓ airflow-worker Pulled                          416.4s
  ✓ airflow-triggerer Pulled                       416.4s
  ✓ airflow-scheduler Pulled                       416.4s
  ✓ airflow-init Pulled                            416.4s
  ✓ redis Pulled                                   165.0s
[+] Running 9/9
  ✓ Network materials_default                      0.8s
  ✓ Volume "materials_postgres-db-volume"         0.2s
  ✓ Container materials-redis-1                   29.2s
  ✓ Container materials-postgres-1                29.1s
  ✓ Container materials-airflow-init-1            113.9s
  ✓ Container materials-airflow-worker-1          115.5s
  ✓ Container materials-airflow-scheduler-1       115.5s
  ✓ Container materials-airflow-webserver-1       118.4s
  ✓ Container materials-airflow-triggerer-1       115.5s
```

Create Admin User



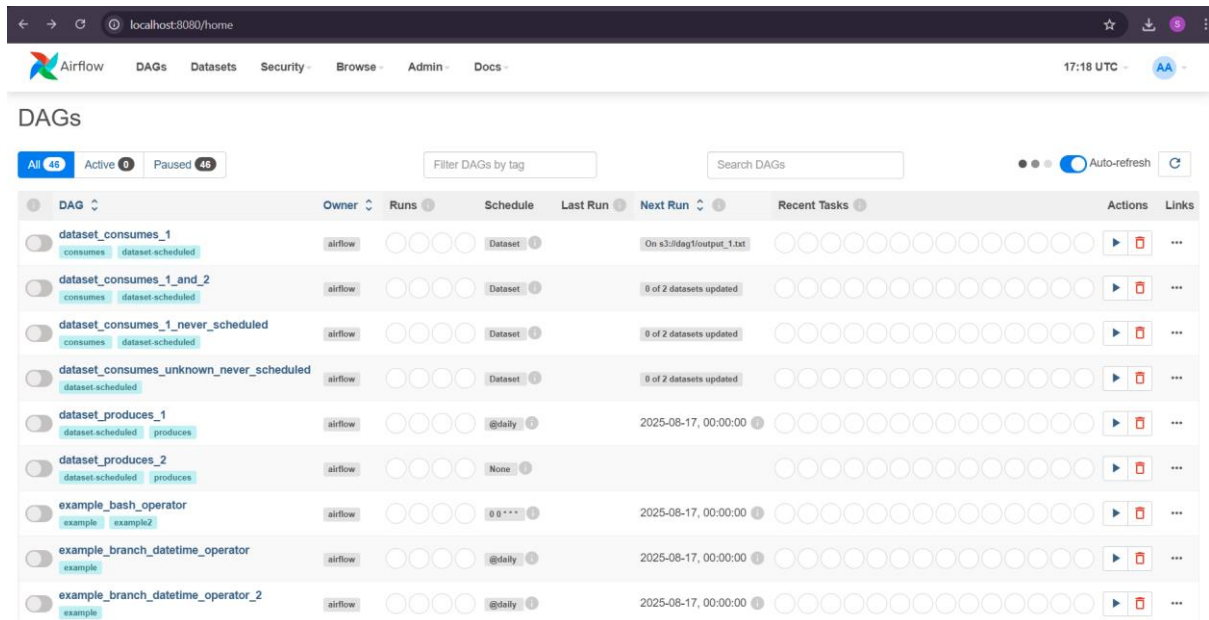
```
PS C:\Users\Dell\Materials> docker-compose run airflow-worker airflow users create --role Admin --username admin --email admin --firstna
me admin --lastname admin --password admin
time="2025-08-18T22:27:45+05:30" level=warning msg="The \"AIRFLOW_UID\" variable is not set. Defaulting to a blank string."
time="2025-08-18T22:27:45+05:30" level=warning msg="The \"AIRFLOW_UID\" variable is not set. Defaulting to a blank string."
time="2025-08-18T22:27:45+05:30" level=warning msg="C:\Users\Dell\Materials\docker-compose.yaml: the attribute `version` is obsolete
, it will be ignored, please remove it to avoid potential confusion"
[+] Creating 3/3
  ✓ Container materials-postgres-1                0.0s
```

Open Airflow Web UI

Open your browser and go to:

<http://localhost:8080>

- Username: admin
- Password: admin



The screenshot shows the Apache Airflow Web UI interface. The top navigation bar includes links for DAGs, Datasets, Security, Browse, Admin, and Docs. The main header displays the Airflow logo, the current time (17:18 UTC), and a user profile icon. Below the header, the 'DAGs' section is active, showing a list of DAGs with filters for 'All' (48), 'Active' (0), and 'Paused' (46). A search bar and a filter by tag input are also present. The DAGs list table includes columns for DAG name, Owner, Runs, Schedule, Last Run, Next Run, Recent Tasks, Actions, and Links. The table lists several DAGs, including 'dataset_consumes_1', 'dataset_consumes_1_and_2', 'dataset_consumes_1_never_scheduled', 'dataset_consumes_unknown_never_scheduled', 'dataset_produces_1', 'dataset_produces_2', 'example_bash_operator', 'example_branch_datetime_operator', and 'example_branch_datetime_operator_2'. Each DAG entry shows its status (e.g., 'On s3://dag1/output_1.txt'), the number of recent tasks, and icons for running, pausing, and deleting the DAG.

DAG	Owner	Runs	Schedule	Last Run	Next Run	Recent Tasks	Actions	Links
dataset_consumes_1	airflow	0	Dataset		On s3://dag1/output_1.txt	0	[Run] [Pause] [Delete]	...
dataset_consumes_1_and_2	airflow	0	Dataset		0 of 2 datasets updated	0	[Run] [Pause] [Delete]	...
dataset_consumes_1_never_scheduled	airflow	0	Dataset		0 of 2 datasets updated	0	[Run] [Pause] [Delete]	...
dataset_consumes_unknown_never_scheduled	airflow	0	Dataset		0 of 2 datasets updated	0	[Run] [Pause] [Delete]	...
dataset_produces_1	airflow	0	@daily		2025-08-17, 00:00:00	0	[Run] [Pause] [Delete]	...
dataset_produces_2	airflow	0	None			0	[Run] [Pause] [Delete]	...
example_bash_operator	airflow	0	0 0 * * *		2025-08-17, 00:00:00	0	[Run] [Pause] [Delete]	...
example_branch_datetime_operator	airflow	0	@daily		2025-08-17, 00:00:00	0	[Run] [Pause] [Delete]	...
example_branch_datetime_operator_2	airflow	0	@daily		2025-08-17, 00:00:00	0	[Run] [Pause] [Delete]	...